

Introduction

FoodNest is a smart food waste reduction application that has been developed as a part of BIT216 Software Engineering Principle assignment at HELP University and this application is designed to address the growing problem of household food waste in Malaysia by providing the users with intelligent inventory management system that have features like expiry tracking, donation facility and weekly meal planning features.

This document presents the Project Plan and the Proposal for the FoodNest application that covers all the requirements component of Task 1 as it outlines in the assignment. The proposal includes a complete comprehensive overview and clearly defined aims and objectives, and a detailed project scope and a full project schedule supported by WBS (Work Breakdown Structure) and Gantt Chart and a technical description of development platform and a risk management plan. This project also follows an iterative and incremental development methodology with two development iterations planned to progressively deliver the six-core use case of FoodNest.

The FoodNest application will be developed as a web-based platform that uses modern technology including React.js for the frontend, Node.js with the Express for the backend and MySQL as the relational database. Our team has chosen a web-based approach to ensure the accessibility across all devices without the need for installation and it makes it easier to demonstrate and deploy within the project timeline and All team members have been assigned the specific use case to ensure the equitable workload distribution and clear individual accountability which will also be reflected in the Gantt Chart WBS and subsequent assignment deliverables.

This project has been undertaken by Group Clovers that consists of three members:

- Nikhil Rana (Team Leader)
- Aaditya Kumar Jaiswal
- Semon Chakhradhar

GitHub Link: https://github.com/LuciFer-Dev69/FoodNest_Software_Engineering_Assignment

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1. Overview

1.1 Management Summary

FoodNest is a smart food waste reduction web application that we will design it to address the growing food waste problem in Malaysian households. We will draw inspiration from the documented issue of excessive food waste in Malaysia, the FoodNest aims to empower users to take control of their food inventory, reduction in unnecessary waste and contribution to more sustainable community through intelligent meal planning and donation facility.

Our web application will be developed as a website platform which includes six core use cases which include user registration with privacy settings, food inventory management, food browsing and donation, food analytics and reporting, notification and weekly meal planning. Our project is undertaken by our Group Clovers, a team of three students from the BIT216 Software Engineering Principles course at Mid Valley International Collage affiliated with HELP University.

And also, The Solid waste Management and Public Cleansing Corporation reported that the food waste in Malaysia stands maximin 9000 tons discarded daily in 2024 that amounts to millions of ringgits in lost resource (BAKAR, 2025).

1.2 Motivation

Food waste is a serious environmental and economic issue in whole world including country like Malaysia. Many households often discard food simply due to poor inventory management, expiry date, and lack of planning. Our website FoodNest is conceived as a practical solution to address these problems at the household level. By providing real time inventory tracking, expiry date alerts and meal planning features, our app will enable users to make smarter decisions about their food consumptions, and this will help to reduce waste and encourage people to handle food more properly and more responsibly.

Also, Malaysian waste approximately has 16.688 tons of food daily that is enough to feed millions of people with nearly 60% of the waste being avoidable and meaning that is consists of food that could have been consumed if managed properly. (Thye, 2025)

1.3 Customer / Target Users

The primary users of FoodNest are primarily targeted to Malaysian household members who wish to manage their food inventory more efficiently, reduce waste, and engage with community donation system and our Secondary users include community volunteers and neighbors who benefit from the donation listings by browsing available food items that are near to them.

1.4 What the Project Will Deliver

FoodNest will deliver a fully functional web application with these following points:

- User registration and privacy/security management.
- Our food inventory management system will track expiry products.
- We will also add food donation and functions like module browsing.

- It will also have visual analytics and dashboard.
- Our system will also provide automated notification for expiry, donation and meal reminders.
- Weekly meal planner on available inventory.

1.5 New Product or Extension

FoodNest will be a new product. While there are also other existing food waste reduction website globally, Our FoodNest will be primary target for Malaysian household context, incorporation donation workflows, dual language consideration and a hyper local community feature that is distinguish from the existing solutions.

1.6 Cost and Duration

Item	Details
Estimated Development Cost	RM 0 (It is an academic project with 0 commercial cost)
Tools & Platforms	Open-Source APIs, Vs Code, Android Studio, GitHub, Gantt Project and Figma.
Project Duration	14 weeks (4 June 2026 – 3 September 2026)
Team Size	3 members (Group: Clovers)

2. Project Aims

Our project aims of the FoodNest project are intentional statements of our team that hopes to achieve upon project completions:

- To design and develop a user-friendly Universal Interface, technology driven web application that assists Malaysian household in reducing food waste through intelligent inventory tracking that will help in households.
- To communicate facility in community engagement by enabling food donation between households and promoting a culture of sharing social responsibility.
- To empower users with data driven insights about their food saving behavior through a proper explained visual analytics and reports.
- To provide a reliable meal planning tool that will help to maximize the user's available food inventory and help users to plan meals around what they have.
- At the end, we ensure that user data security and privacy through robust and strict authentication mechanisms including 2FA and MFA.

3. Project Objectives

The following measurable objectives define the specific outcomes the project will accomplish:

1. **Deliver a working user registration and privacy module:** Use Case 1 with 2FA support will end by 28 July 2026.
2. **Implement a food inventory management system:** Use Case 2 that allows users to log, edit and track food items with expiry dates and other functions like storage categories.
3. **Develop a browse and donation interface:** Use Case 3 enables users to search, filter, and claim the donated food in the community.
4. **Build a food analytics and reporting dashboard:** Use Case 4 that displays visual progress like indicators and historical data on food saved and donations that are made.
5. **Implement an automated notification system:** Use Case 5 will help to alert users on expiry dates, donation status and meal planning reminders.
6. **Create a weekly meal planner:** Use Case 6 will link the inventory items to planned meal, and the schedules reminder will be based on the ingredient that is available.

7. **Complete and submit all project documentation:** As the academic requirement by the specified deadline will be done in time (Task 1: 12 June 2026).

4. Project Scope

4.1 In Scope

The following features and deliverables are within the scope of the FoodNest project:

- User account registration, login and privacy and security setting that include 2FA and MFA configurations. – Use Case 1
- Food inventory management like adding, editing, deleting and category food items with expiry tracking. – Use Case 2
- Food browsing and donation system like posting, browsing, filtering and claiming food donations. – User Case 3
- Food analytics like visual dashboard like displaying food saved, waste reduction and donations. – Use Case 4
- Notification system like automated alert for expiry, donation updated and meal reminders. – Use Case 5
- Weekly meal planners like planning meals for existing inventory with system suggested recipes. – Use Case 6
- Deployment in GitHub with a public repository that will be maintained by team members.
- Full project documentation like proposal, requirements specifications and design document.

4.2 Out of Scope

Here are the scopes for the FoodNest project:

- Integration of free open-source stores APIs.
- Monetization and Payment features.
- Use of Advance AI/ML recipe's generations.
- Integration of Physical hardware.
- Multi-language support.
- Commercial deployment on android store (Play Store).

5. Project Schedule

5.1 Work Breakdown Structure (WBS)

Our WBS if FoodNest project we have distributed in 6 phases, 29 tasks and 9 milestones/deliverables across in 14-week timeline. And Each task is assigned to a responsible team member that includes the duration in working days, predecessor dependencies and classification as a milestone or deliverable where it is applicable.

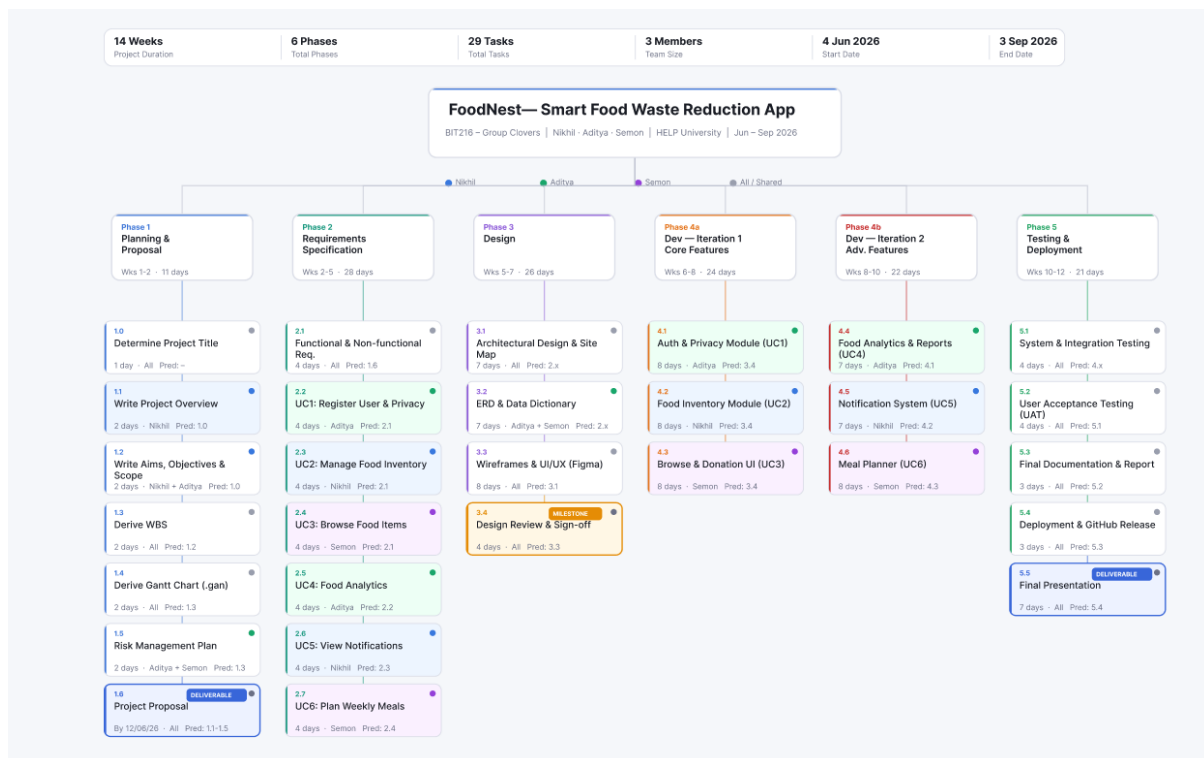


Figure 1: WBS Diagram

Phase 1 – Planning & Proposal (Weeks 1-2, 11 days):

- The project foundations will include scope, objectives, team members structure and planning of artifacts.
- Task Distribution and Discussion: Project Overview, write aims, Objectives and Scopes, Derive WBS, Derive of Grantt Chartt, and Risk Management Plan.
- Deliverables: 1.6 Project Proposal submitted 12 Jun 2026. Lead by Nikhil.

Phase 2 – Requirements (Weeks 2-5, 28 days):

- Documentation of all functional and non-functional requirements with the dedicated use case specifications for each of the 6 core features.
- Task: All Functional and Non-Functional Requirements, Use Case 1 – 6 specifications.
- Task Distribution: Aditya (UC1, UC4), Nikhil (UC2, UC5) and Semon (UC3, UC6).

Phase 3 – Design (Weeks 5-7, 26 days):

- It will cover all the system and UI/UX design before development begins.
- Task: Architectural Design and Site Mapping, ER Diagram and Data Dictionary, Wireframes and UI/UX in Figma.
- Milestones: 3.4 Design Review and Sign off. Internal checkpoints before any coding phase starts.
- Task Distributions: All members as Aditya and Semon will be leading ERD.

Phase 4a – Development: Core Features (Weeks 6-8, 24 days):

- We will start First Development Sprint where all three main features start at same time.
- Task: Auth and Privacy Module (Use Case 1), Food Inventory Modules (Use Case 2), and Browse and Donation UI (Use Case 3).
- Task Distribution: Aditya (Use Case 4), Nikhil (Use Case 2), and Semon (Use Case 3).

Phase 4b – Development: Advance Features (Weeks 8-10, 22 days):

- Second Development sprint will be starting where three advance use cases on will begin for top of the core system.
- Task: Food Analytics and Report (Use Case 4), Notification System (Use Case 5) and Meal Planner (Use Case 6).
- Task Distribution: Aditya (Use Case 4), Nikhil (Use Case 5), Semon (Use Case 6).

Phase 5 – Testing and Deployment (Weeks 10-12, 21 days):

- Team will validate full system, and team will prepare for final submissions.
- Task: System and Integration Testing. User Acceptance Testing, Final Documentation and Deployment & GitHub release.
- Deliverable: 5.5 Final Presentation.
- Team Distribution: All members.

Team Roles and Color Coding:

Member	Roles	Color
Nikhil Rana	Team Leader, Project Planning, UC2 and UC5.	Blue
Aditya Kumar Jaiswal	Requirements, UC1, UC4.	Green
Semon Chakhradhar	UC3, UC6, ERD	Purple
All	Collaborative tasks	White

5.2 Milestones & Deliverables

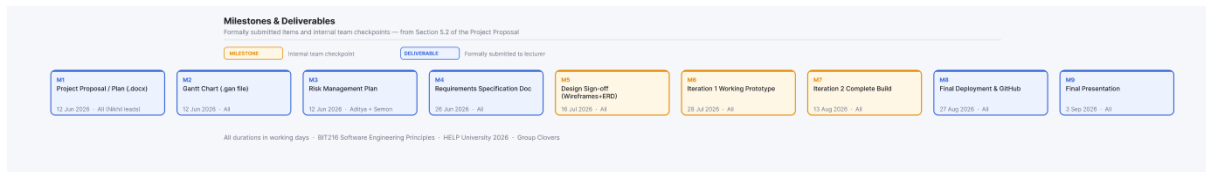


Figure 2: Milestone and Deliverables

Here is the Milestone and Deliverables table:

#	Types	Name	Date
M1	DELIVERABLES	Project Proposal and Plan	12 Jun 2026
M2	DELIVERABLES	Gantt Chart	12 Jun 2026
M3	DELIVERABLES	Risk Management Plan	12 Jun 2026
M4	DELIVERABLES	Requirements Specification Docs.	26 Jun 2026
M5	MILESTONES	Design (Wireframe and ERD)	16 Jul 2026
M6	MILESTONES	Iteration 1	28 Jul 2026
M7	MILESTONES	Iteration 2	13 Aug 2026
M8	DELIVERABLES	Final Deployment and GitHub Release	27 Aug 2026
M9	DELIVERABLES	Final Presentation	3 Sep 2026

5.3 Gantt Chart

The Gantt Chart was created using application named Gantt Project version 3.3. We chose this application because it captures all phases, tasks, predecessors, resource allocations, and milestones dates for the FoodNest Project. This Gantt chart spans from 4 June 2026 to the month of September 2026 across 14 weeks.

The following screenshot shows the full Gantt Chart view, including all phases, task timelines, and team member allocations:

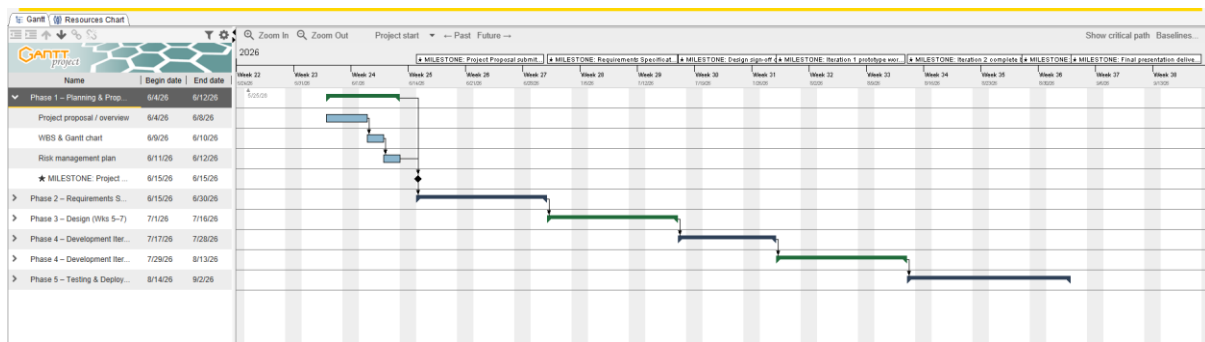


Figure 3: Gantt Chart Phase I

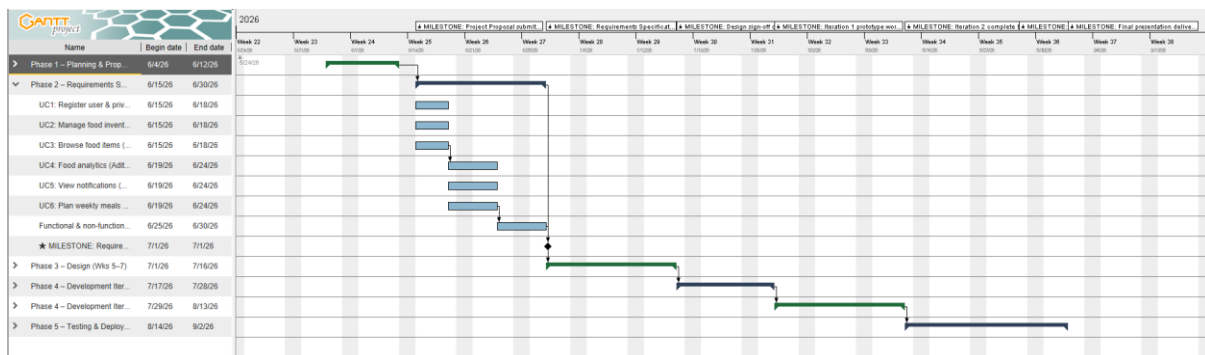


Figure 4: Gantt Chart Phase II

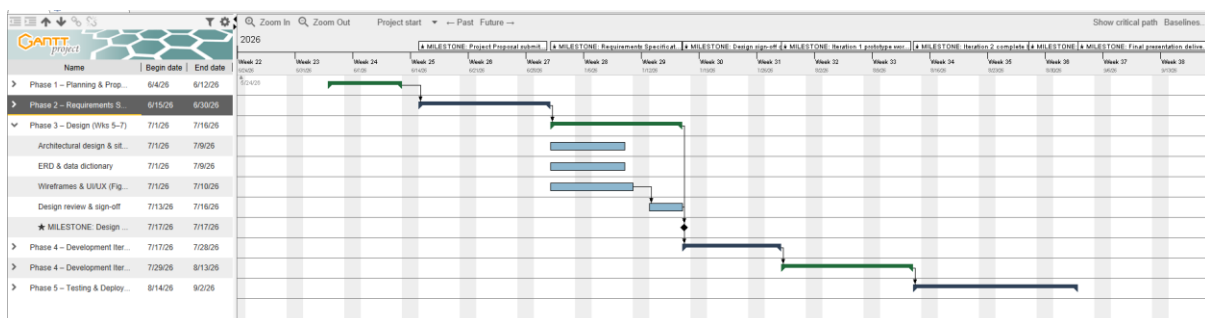


Figure 5: Gantt Chart Phase III

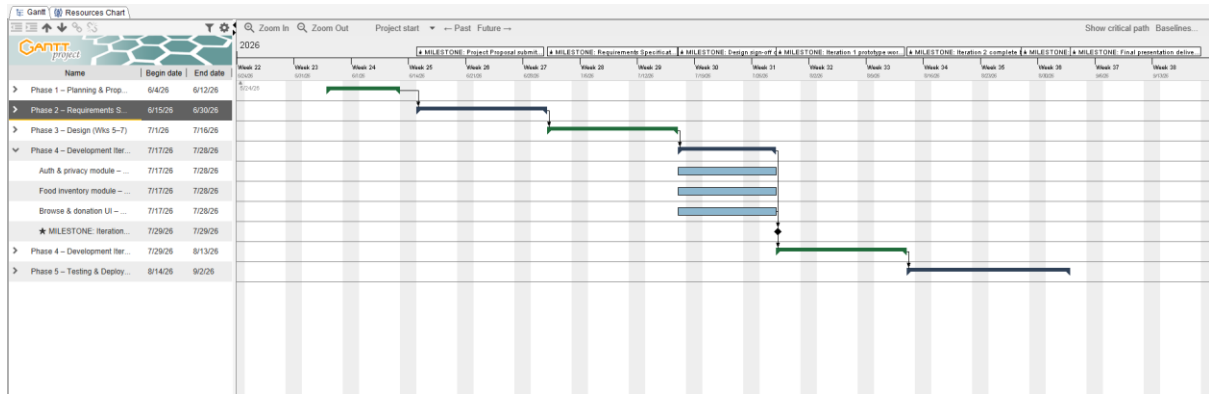


Figure 6: Gantt Chart Phase Iv a

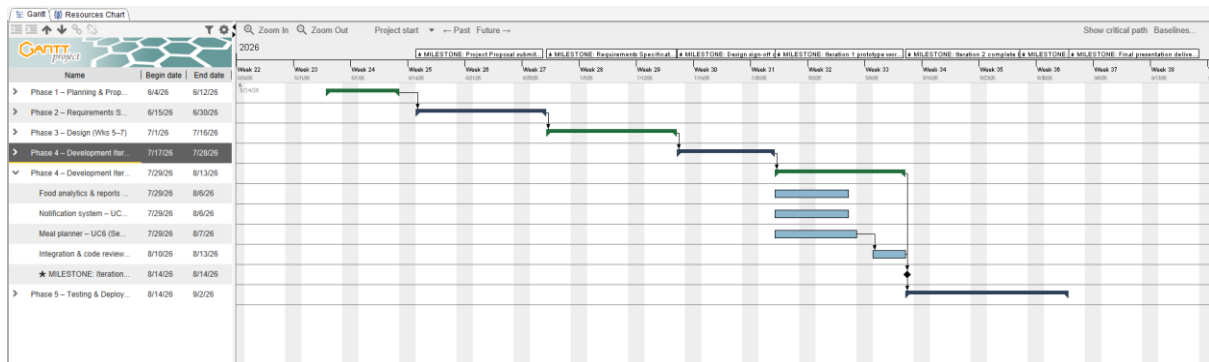


Figure 7: Gantt Chart Phase Iv b

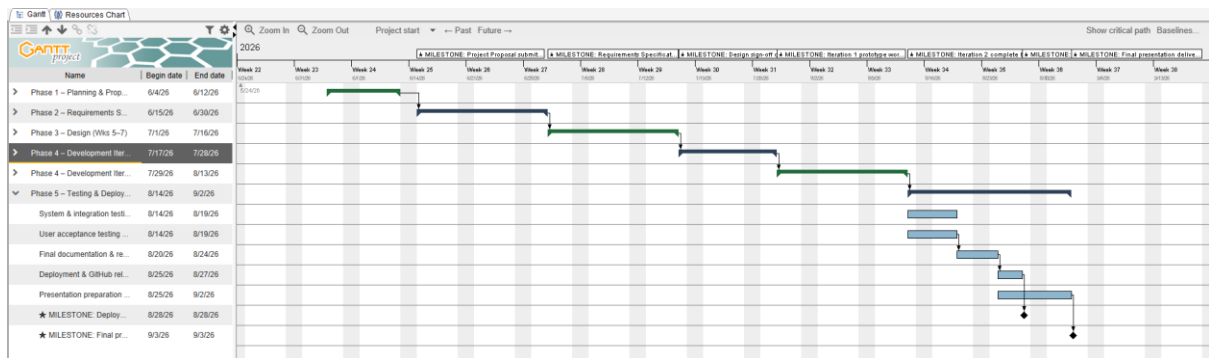


Figure 8: Gantt Chart Phase V

Resource Chart:

Here is the resource chart in Gantt Project that displays the workload distribution across all three team members Nikhil Rana, Aditya Kumar Jaiswal and Semon Chakhradhar. Throughout the project, June to September 2026. And each colored bar will represent the assigned task per member per week.

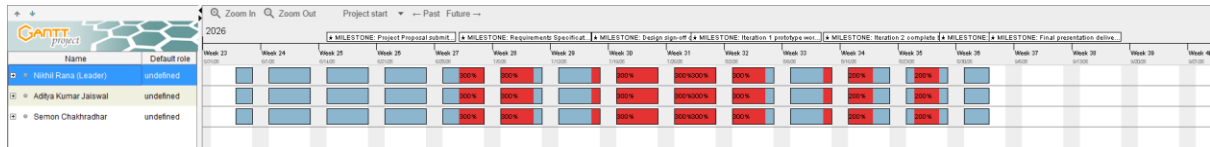


Figure 9: Resource chart

In figure 9, the cell highlighted in the red indicates week where a team members workload is exceed to 100 percent capacity, and particularly during week 28 – 31 of development in Iteration 1 and 2 where all of the three members are simultaneously working on their respective use case modules and the Cell in blue will represent a normal work load weeks and the chart confirm that each members carries a balanced and comparable overall load across the 14 week project with the peak intensity during the development phase and reduced load during testing and closure phase.

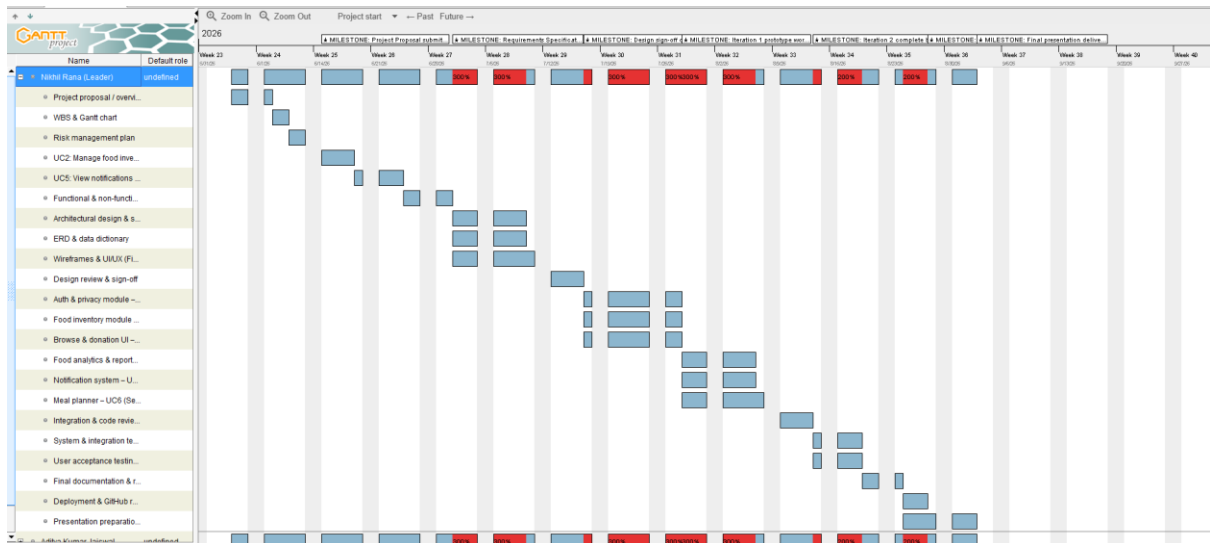


Figure 10: Resource Chart (Nikhil Rana Leader)

This resource chart is for Nikhil Rana where it shows the consistent task allocation across the full 14 week project durations where Nikhil will lead Planning and Proposal phase where it is followed by requirement tasks and contributes to all shared Design and Testing phase.

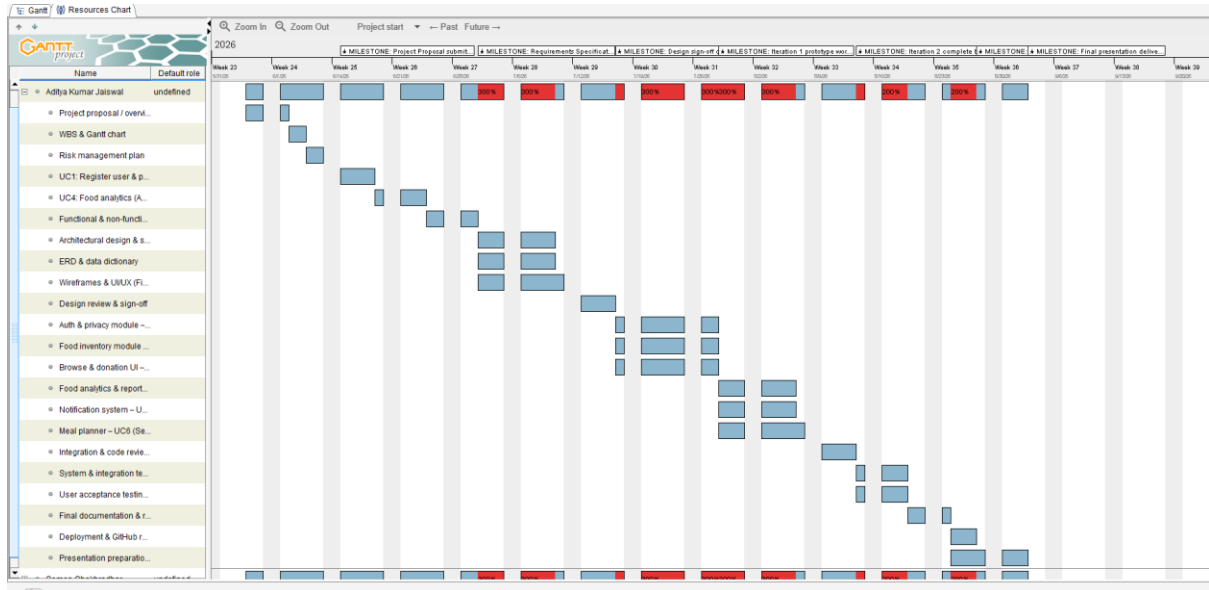


Figure 11: Resource Chart (Aditya Kumar Jaiswal)

This is the resource chart of Aditya Kumar Jaiswal where this chart will reflect active involvement from the early planning phase throughout the whole project closure and his primary ownership covers his UC1 and UC4 along the shared collaboration to Risk Management, Requirements and Design Task.

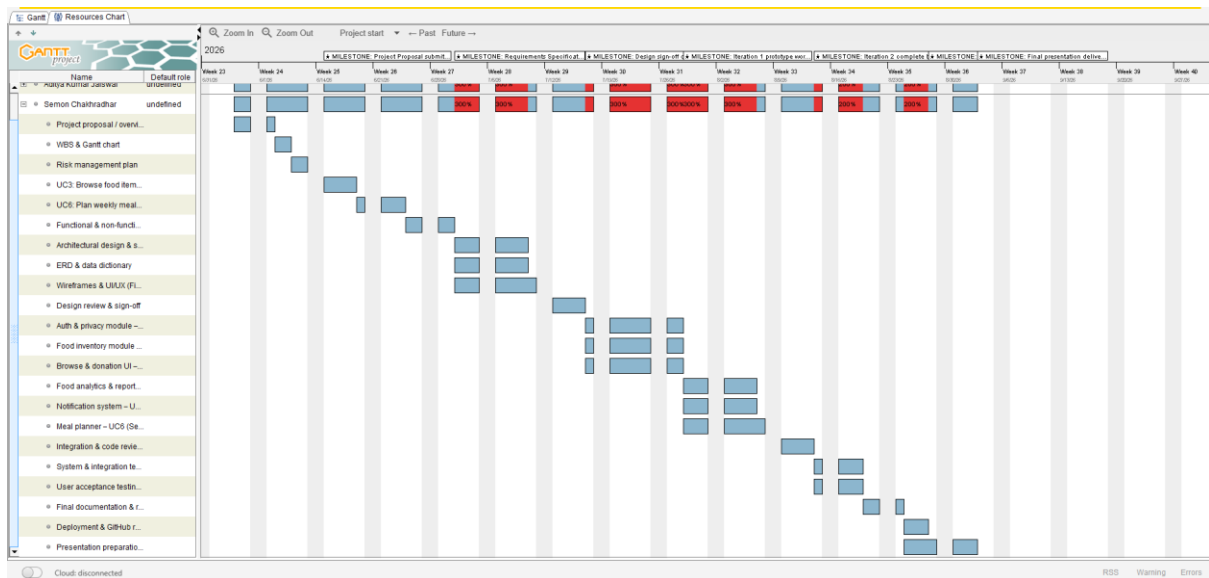


Figure 12: Resource Chart (Semon Chakhradhar)

This is the Semon resource chart which follows the same overall pattern as the team with the early involvement in Planning and Risk Management, primary ownership of UC3 and UC6 and shared responsibility across Design, Integration and Testing phase.

6. Technical Description / Development Platform

6.1 Overview of Technical Approach

FoodNest will be developed as a web-based application that will be using full-stack development and architecture. Our frontend will provide a responsive user interface accessible across device and while our backend will handle all of the important things like business logic, data persistence and notification management. This project will be a follow of iterative and incremental development methodologies as a defined in the project schedule.

6.2 Development Platform

Documentation & Modelling

Criteria	Tool / Software	Version	Description
Documentation	Microsoft Word	2019/365	It will be used to write all of the format of all project documentation including proposal, requirements spec and used to design documents.
Modelling – Req.	Draw.io / Lucidchart	Free tier	It will be used to create Use Case Diagram as well as system flow models.
Modelling – Design	Draw.io	Free tier	It will be used to create ERD, Data Dictionary diagram and Architecture Diagrams.
UI Prototyping	Figma	Free tier	It will be used to design our website wireframes and user interface mockups for all of the use case.
Project Management	GanttProject	3.3.3322	It will be used to create and manage the project like Gantt Chart and resource allocations.
Collaboration	GitHub	Latest	We will use GitHub for version control and team collaboration repository and to host all source code and documentation.
Communication	WhatsApp / Discord	Latest	It will be used for communication, meeting coordination and progress updates.

Framework & Development

Criteria	Tool / Software	Version	Description
Frontend	HTML5, CSS3, JavaScript / React.js	React 18+	We will use this modern framework for frontend and it will be responsible for responsive web UI built with component

			based React framework for the process of data rendering.
Backend	Node.js with Express.js	Node 20 LTS	For Backend the RESTful API server will be handling all of the business, logics, authentications and database operations.
Database	MySQL / MariaDB	8.0+	Relational Database will be used for storing user account, food inventory, donations, meal plans and for analytics data.
Authentication	JWT + bcrypt + TOTP (2FA)	Latest	JSON web tokens will be responsible for authentication like session management, b-crypt for password hashing and we will use TOTP for 2-factor authentications.
Notification	NodeMailer / Web Push API	Latest	Our web application will also have function like email notifications for expiry alerts; and browser push notifications for the real time updates.
Testing	Jest + Selenium	Latest	For Testing we will use unit and integrations like testing with Jest; functional / UI testing with using Selenium Web Driver.

Hardware Specifications (Development Notebook)

Component	Specification
Processor	Intel Core i5 / AMD Ryzen 5 (or equivalent) minimum quad-core
RAM	8 GB minimum (16 GB recommended for running local dev server + database)
Storage	256 GB SSD minimum (for project files, node_modules, and database storage)
Operating System	Windows 10/11, macOS 12+, or Ubuntu 20.04 LTS and Other Linux
Display	13-inch minimum for development; Full HD (1920×1080) recommended.
Internet Connection	Stable broadband will be required for GitHub pushes and online collaboration.

Demonstration Platform

Criteria	Tool	Version	Description
Browser	Google Chrome	Latest stable	Primary demo browser, supports all HTML5, CSS3, and JavaScript will be used in the app.
Browser (Alt)	Mozilla Firefox	Latest stable	Be used by Fallback Demo browser for cross browser compatibility verifications.

Local Server	localhost (Node.js)	Port 3000	Application will be run locally for this development machine during the demonstrations.
Database	MySQL Workbench	8.0+	It will be used to demonstrate data persistence and for query execution during UAT.

7. Risk Management Plan

The following risk management tables will show the FoodNest project that identifies the possible risks and checks how likely they are to happen and how serious their effect could be and creates plan to prevent them and handle them if they occur as a team.

Risk Name	Description	Prob.	Mitigation Strategy
Team Member Unavailability	Team Member may be not present due to illness, personal problem or any other academic pressure that causes delay to assigned tasks.	M	There will be redistribution of task immediately among the remaining members and there will be shared documentation on GitHub so any member can continue another member work.
Scope Creep	There might be Additional feature that may be requested or assumed during the development, expanding the project beyond its defined boundaries that might cause schedule overrun.	M	We will stick to 6 agreed features and if any changes team need it will be discussed with team leader's approval. Then, we will finalize and lock the project scope.
Technical Difficulties	Team members may encounter unfamiliarity with any tools and frameworks like React, Node.js or MySQL leading to Implement delays.	H	Members will use the first week of each iteration for learning and project setup and use resources like YouTube, Stack Overflow and official documentation for support and seeking help from the lecturer.
GitHub Conflicts / Code Loss	Continuation of edits by multiple teams might cause problem merge conflicts or accidental file overrides that might result in lost work.	M	We will enforce a branch strategy where each member will work on a separate branch and submits pull request for review before merging to main and Commit changes at least once daily.
Deadline Overrun	There might be delays in task and may take longer than estimated that might cause the team to miss interim or final submission deadlines.	M	Each member will have a 2-day build buffer before major deadlines that track progress weekly and using the Gantt Chart. And inform the team leader if any task is delayed by more than one day.
Communication Breakdown	Poor communication between team members may cause duplication of work, misaligned expectations or decisions.	L	There will be team meetings online or in-person using a application like WhatsApp and Discord groups for quick updates and Document all of the key decision in a shared notes file in GitHub.

Data Loss (Local Files)	Local Hardware failure like laptop crash could result in loss in project files, documentation or source code.	L	All Team members must push their code/document to GitHub at the end of their work sessions or any Critical documentation should be backed up in Online cloud storage services like Google drive, One Drive and Proton Drive etc.
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References

- BAKAR, N. N. (2025, 4 10). Retrieved from Malaysian Reserve:
<https://themalaysianreserve.com/2025/04/10/malaysias-ramadhan-food-waste-solutions-in-action/>
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CONCLUSION

In conclusion, we would like to say that this project plan document has presented as a comprehensive and well structured plan for the development of FoodNest a smart food waste reduction web application for Malaysian household and this document has clearly outlined as the project motivation, aims, objective and scope this will help to established a solid foundation for the development work that will follow in the subsequent phase. This project has six well defined use cases that cover user registration, food inventory management, donation browsing, food analytics, notification and meal planning for the FoodNest application that is designed to deliver meaningful and practical value to the user targets.

This project schedule reflect in the Work Breakdown Structure and Gantt chart, demonstrates a realistic and achievable 14 weeks' timeline that follows an iterative and incremental development methodology and the task has been distributed equitably among the three members with each members taking ownership of specific use case across both development iteration and the risk management plan further ensure that the team is prepared to handle potential challenge together.

At the end I would like today that Group Clovers is confident that the FoodNest project will be planned as in this proposal and will be executed in a disciplined and professional manner and in accordance with the principles of software engineering. The Clover team is committed to this meeting and all submission deadlines that maintain clear communication and collaboration throughout this project life cycle and producing a final product that fully addresses the requirements that are set in the case study. This proposal serves as the guided document for all the future development, design and testing activities that will be carried out in these coming weeks.